

BOWEN JIANG

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EDUCATION

Duke University

B.S. in Biomedical Engineering and Electrical & Computer Engineering

Durham, NC

Expected May 2027

- **GPA:** 3.9/4.0 | **Coursework:** Deep Neural Nets, Neuromodulation, Signals & Systems, Microelectronic Devices, Data Structures, Quantitative Physiology & Biostats, Linear Algebra, Computer Architecture, Biochip Engineering, Modern Diagnostic Imaging Systems

RESEARCH & INTERNSHIP EXPERIENCE

Tetsuwan Scientific

Lab Automation Intern

San Francisco, CA

May 2025 – Aug 2025

- Automated Nanopore sequencing and bacterial plating pipelines on Opentrons Flex, reducing hands-on time by 90%.
- Built an OpenCV-based colony-picking system with 95% recognition accuracy and 3× faster plate processing.
- Launched LabAutoWiki.org, the go-to site for lab automation information, scaling to 12+ contributors and 50+ pages.

Cogan Lab

Student Researcher

Durham, NC

Sep 2025 – Present

- Built MNE pipelines from response coding, high gamma extraction to significant electrode visualization for speech perception decoding; individually analyzed 24 patients' sEEG data to understand the uniqueness points of words based on their phonemes.

Chory Lab

Student Researcher

Durham, NC

Aug 2024 – Present

- Automated T7 phage-based directed evolution workflows for chromatin-binding protein engineering on Hamilton.
- Optimized PCR and Qubit reaction cost by > 50% while increasing throughput on the Opentron OT-2; contributed visualizer modules to PyLabRobot; documented automation protocols and curated code base for future use.

Barrett Lab at the Broad Institute

Student Researcher

Cambridge, MA

Jun 2024 – Aug 2024

- Conducted Cell Painting Assay and induced iPSC-derived neurons & progenitor cells for Down Syndrome morphology observations.
- Applied ML with XGBoost to extract key traits related to DS (nuclei area, signal intensity, etc.), reached 83% prediction accuracy.

Apple

Summer Data Scientist, Apple DMP Team

Beijing, CN

Jun 2023 – Aug 2023

- Constructed a Prophet-based run-rate forecasting model to predict 15-week Mac sales, capturing weekly/quarterly seasonality, holiday effects, and new product launch; reduced MAPE by 34% vs. legacy ES model and productionized the pipeline to data team's workflow.
- Built dealer- and POS-level features (run-rate ratios, pricing deltas, etc.) with Pandas and evaluated LightGBM models to forecast weekly iPhone sales, enabling quantitative validation of distributor performance by isolating true demands from channel-driven noise.

SIDE PROJECTS & LEADERSHIP

TryIt

Co-Founder

San Francisco, CA

Jul 2025 – Present

- Built a personalized fashion shopping SaaS that supports prompt-based virtual try-on; leverage Gemini and OpenAI APIs to deliver realistic outfit selection on avatar images; collaborated with the Duke University Store to bring the product live.

CalendarOCR

Founder

Durham, NC

Sep 2025 – Present

- Shipped an iOS app for poster-to-calendar conversion using OCR + LLM; 100 downloads, 10.2% conversion, 2% paying conversion.

Duke Chinse Student Association

Co-President

Durham, NC

Apr 2025 – Present

- Lead the largest undergraduate Chinese cultural organization with 10+ annual events and 50+ active members.

PUBLISHED/SUBMITTED PUBLICATIONS

- "Morphological Differences of Neural Progenitor Cells in Down Syndrome", *HSURV Abstract Book*, September 2024.
- "Discovery of Kidney Disease Targets Using Multimodal Human Podocyte Injury Model", Manuscript submitted to *Stem Cell Reports*, June 2025.

SKILLS & INTERESTS

Skills: Python (Pandas, scikit-learn, PyTorch), Swift, Java, C, C++, CAD, HTML, OpenCV, PLR, Time-series Model, R, MNE, BrainSuite

Familiar Models: ESM2, DDPM & DDIM Diffusion Models, YOLOv8, Prophet, LightGMB, U-Net, ResNet, GPT